
Structured Slot Memory for Masked-Evidence World-Model Interfaces

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Paper under double-blind review

Abstract

Finite-context world models can observe a decisive visual cue and later act as if it was never seen. We study this failure as an interface problem: old evidence must be stored beyond the raw context window and must also be exposed through a frozen host representation in a form a later consumer can use. We introduce a self-supervised masked-evidence JEPA objective with a compact slot-memory writer. Training never receives cue labels, cue crops, or manual key-frame annotations. Instead, salient target frames and patch sets are mined from the rendered stream, withheld from the legal context, and predicted from a causal memory state. On twelve OGBench decks, three controlled evidence ages (4, 8, and 15), and three seeds, the slot-memory method passes every registered retention cell: mean full-memory balanced accuracy is 0.983, while reset and no-state controls remain at 0.250 and 0.248. Under the same data, targets, loss, and readout, GRU, LSTM, and Mamba-lite baselines reach 0.803, 0.577, and 0.800 mean full-memory balanced accuracy and pass 22/36, 6/36, and 22/36 environment-age rows. The age-15 stress test is the clearest separator: slot memory reaches 0.962, versus 0.669, 0.350, and 0.683 for the baselines. For supported environments, fixed-controller native-use checks score 27/36 environment-age rows with 0.957 mean executed success. We scope the claim carefully: this is evidence for structured old-evidence retention and exposure under controlled ages, not autonomous planning or automatic memory-age selection.

1 Introduction

Finite-context world models often fail in a way that is easy to miss: they can observe the decisive cue early and later produce a representation that behaves as if the cue was never present. This matters for physical intelligence because action-relevant evidence is rarely synchronized with the moment of action. A robot can read a sign before turning away, see a door state before entering a hallway, or observe an object identity before the object becomes occluded. If the interface to the frozen host model exposes only a recent window, then locally plausible prediction is not enough; the old evidence must survive long enough to affect the later decision.

This paper studies that failure as an interface problem. We do not ask only whether a hidden state can memorize something. The stricter question is whether old visual evidence, after it has left the legal context window, is retained in memory, exposed through a host-facing representation, and usable by a bounded downstream controller. This framing separates three claims that are often conflated. A model may store old evidence internally but fail to expose it. A readout may decode a label from a shortcut if the current endpoint still leaks the cue. A native controller may succeed because it received the true goal rather than the memory-selected one. Our evaluation is designed to close these loopholes.

We introduce a masked-evidence slot-memory interface. Training is self-supervised: it never receives cue labels, cue crops, or manually selected key frames. Instead, the training pipeline mines visually salient patch sets from the rendered causal stream, masks those patches from the visible source, and asks a compact slot transformer to predict their stop-gradient latent

targets. Cue labels enter only after training, when a post-hoc audit readout tests whether the learned memory exposes the old cue under matched full, reset, and no-state conditions.

The central empirical point is structural rather than just numerical. Under the same caches, target mining, patch target encoder, loss, optimizer, and readout, replacing the slot writer with GRU, LSTM, or Mamba-lite recurrence weakens long-delay exposure. The baselines can often solve short waypoint-like rows, which is useful evidence that the task is not impossible. Their failure appears when the cue is old or distributed across object-like visual evidence. Slots help because the target itself is a set of visual facts; a memory set is a better carrier for that target than a single hidden vector.

Our contributions are: a fail-closed audit for old-evidence memory in frozen world-model interfaces; a label-free masked patch-set objective for training a slot memory writer; a controlled architecture comparison across twelve OGBench render decks, three evidence ages, and three seeds; and a bounded native-use check showing that the exposed memory changes target selection where audited fixed controllers are available.

2 Related Work

World models and finite context. Latent world models compress observations into predictive state and use that state for control or planning (Hafner et al., 2023). This compression is powerful but makes old evidence a first-class interface concern: the host can produce reasonable short-horizon predictions while dropping information needed for a later choice. OGBench supplies diverse offline goal-conditioned environments (Park et al., 2024); we use rendered OGBench decks as a controlled substrate for old-evidence tests rather than as an end-to-end planner benchmark.

Self-supervised prediction. JEPA-style objectives predict hidden representations instead of pixels (Assran et al., 2023), and contrastive objectives such as InfoNCE encourage a model to distinguish the positive target from alternatives (Oord et al., 2018). Our use case is narrower: the target is not the next frame, but visual evidence that has been deliberately removed from the legal context. The target miner is generic and label-free, so the audit can later ask whether semantic cue information emerged without being trained as a class label.

Memory architectures. GRU and LSTM recurrence (Cho et al., 2014; Hochreiter and Schmidhuber, 1997) provide classic sequence memory, and selective state-space models such as Mamba (Gu and Dao, 2023) provide a stronger long-sequence prior. Transformer attention (Vaswani et al., 2017) and slot-based object representations (Locatello et al., 2020) motivate a different view: memory should be a structured set when the evidence itself is multi-part. Our experiments keep the objective fixed and vary the writer, so the comparison probes the memory interface rather than the supervision signal.

3 Preliminaries

Let $x_{0:T}$ denote rendered frames and $a_{0:T-1}$ actions. A cue c is visible at an early time and determines a later target label y . At readout time t , the frozen host sees only a recent context window of length K ; the cue is outside that window. We call the delay between cue visibility and the readout boundary the evidence age. The current registered ages are 4, 8, and 15, which act as controlled stress tests rather than automatically discovered memory horizons.

The audit has three conditions. The full condition uses the causal memory state M_t . The reset condition observes the same endpoint and recent context but clears persistent memory at the context boundary. The no-state condition removes persistent memory entirely. A cell passes only when full-memory balanced accuracy is high while both controls remain near chance. This fail-closed rule is the important part: a high full score is not accepted if the endpoint or recent context can solve the task.

We additionally run a bounded native-use check where a local audited controller exists. The memory readout selects a target; the controller receives that selected target rather than the true label; the environment reports success against the original target. This is not autonomous planning. It is a controlled test that exposed memory changes a downstream physical outcome.

Table 1: Claim ladder used throughout the paper. Each gate removes a common shortcut before the next, stronger claim is considered.

Gate	What must be true	What the gate rules out
Demand	The decisive cue is outside the legal recent context at readout time.	Solving the task from the endpoint or a short observation window.
Retention	Full memory exposes the old cue on held-out episodes.	A memory objective that reconstructs generic texture but loses the cue.
Controls	Reset and no-state conditions remain near chance under the same readout.	Leakage through actions, endpoint pixels, class imbalance, or readout bias.
Use	A fixed controller succeeds when driven by the memory-selected target.	A representation that is decodable but irrelevant to execution.

Table 1 keeps the paper’s claim honest. The first three gates are necessary for every environment-age cell. The fourth gate is reported only when the repository has a local controller that can be audited independently. This is why missing native-use rows are not silently counted as successes. They are marked as controller-unavailable and remain outside the native-use claim.

4 Methodology

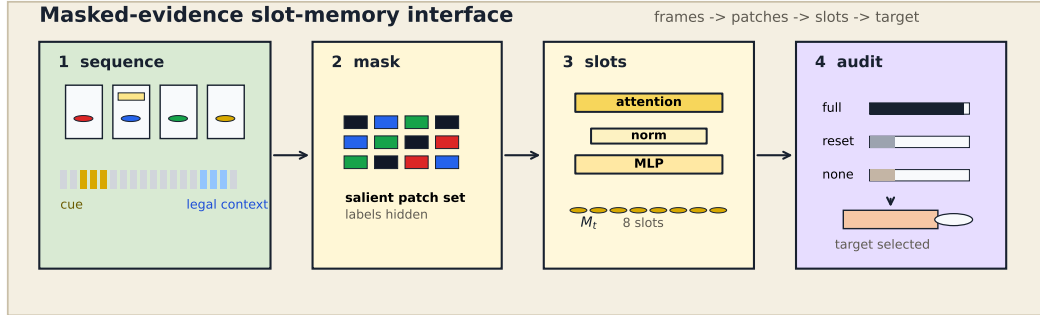


Figure 1: Visual-first architecture. The cue appears in the rendered sequence, target patches are mined and masked without labels, a compact slot transformer stores the old evidence, and the audit checks whether the memory-selected target survives full/reset/no-state contrasts.

Figure 1 summarizes the method as the paper uses it. The left side is the physical stream: frames, actions, and time indices arrive causally. The target side is built from the same stream but is hidden from the visible source. The middle is the learned memory writer. The right side is the audit, which is intentionally outside the training loss.

Table 2: Module contract. The training path is label-free; labels appear only in the post-training audit path.

Module	Input and output	Design reason
Trajectory cache	Rendered frames, actions, time indices, and split metadata.	Keeps all methods on the same stream and validation rows.
Patch miner	Selects salient target patches from the causal history without labels.	Prevents a manual cue crop from becoming the supervision signal.
Slot writer	Maps visible tokens into a small persistent slot set M_t .	Gives distributed old evidence multiple carriers instead of one vector.
Set predictor	Predicts unordered target-patch latents with InfoNCE and cosine matching.	Matches the multi-patch target while avoiding a hand-coded slot order.
Audit readout	Fits labels after training and evaluates full/reset/no-state features.	Measures exposed old evidence rather than the self-supervised loss alone.

The module contract clarifies why the comparison to recurrent baselines is meaningful. GRU, LSTM, and Mamba-lite are not trained on weaker data. They receive the same mined targets and are judged by the same readout gate. The difference is the memory format used to carry old evidence from the causal history to the host-facing representation.

Masked patch-set target. For each episode, the target miner divides the causal history into temporal bins and selects visually salient, novel patches from those bins. The miner does not know cue labels, cue positions, or task-specific object crops. The selected patches are removed from the visible stream and encoded by a fixed target encoder into color and texture latents. This design makes the training signal old and visual without making it semantic supervision.

Slot writer. The writer receives frame tokens, action tokens, and time tokens. A compact slot transformer updates a fixed number of normalized memory slots. Prediction is set-valued: an MLP maps slots to target-patch latents, and the loss combines set InfoNCE, cosine matching, and a small anti-collapse standard-deviation regularizer. Because both the memory and target are sets, the model is not forced to squeeze every old fact into one recurrent vector or one hand-indexed slot.

Shared audit readout. After training, a ridge readout is fitted on train memory features and cue labels. The same readout is then applied to held-out full, reset, and no-state features. Labels are therefore read only for evaluation, not for optimizing the memory writer. The baselines replace the slot writer with GRU, LSTM, or a plain PyTorch Mamba-lite recurrence, but keep the target miner, target encoder, loss, features, and readout contract unchanged.

The training and inference procedures are deliberately asymmetric. During training, the model only receives reconstruction pressure on hidden patch sets. During evaluation, the readout asks a semantic question that was not optimized directly. This asymmetry is important: if labels were used during training, a reviewer could attribute success to supervised cue classification. If labels were never used at all, there would be no way to audit whether the memory retained the task-relevant cue. The split between label-free training and label-revealed evaluation is therefore a measurement design, not a loophole.

5 Experimental Results

Protocol. We evaluate twelve rendered OGBench decks: PointMaze medium, large, giant, and teleport; AntMaze large and giant; HumanoidMaze large; Cube single, double, and triple; Puzzle-3x3; and Scene. Every memory writer is run at the same three evidence ages and three seeds. The result artifacts used by this paper are generated from local JSON and NPZ outputs by scripts/plot_paper_c.py.

Table 3: Controlled-age retention and exposure on 12 OGBench decks, three ages, and three seeds. Full, reset, and no-state are balanced accuracies from the post-hoc audit readout.

Method	Full	Reset	No-state	Seed pass	Env-age pass
Slot memory (ours)	0.983	0.250	0.248	108/108	36/36
GRU	0.803	0.252	0.250	73/108	22/36
LSTM	0.577	0.248	0.250	26/108	6/36
Mamba-lite	0.800	0.255	0.253	71/108	22/36

Table 3 is the compact scorecard, but the interpretation is not that a single average proves memory. The full/reset/no-state pattern is what matters. The slot model exposes old evidence while the matched controls stay close to chance, so the readout is not merely exploiting endpoint color, action sequence, or short-window context. The recurrent baselines are useful negative controls: they share the same self-supervised target and can solve a large fraction of easier cells, yet they lose the consistent exposure pattern on older and more structured evidence.

Two details make this comparison stronger than a raw leaderboard. First, the controls are matched at the feature level: full, reset, and no-state readouts are evaluated on the same held-out episodes. A method cannot pass by making every endpoint easy, because the controls would pass as well. Second, the baselines are strong enough to reveal where the task changes character. Their short-age successes show that the target miner is learnable and that the readout is not artificially hostile. Their long-age failures show that the memory interface, not only the loss, is doing work.

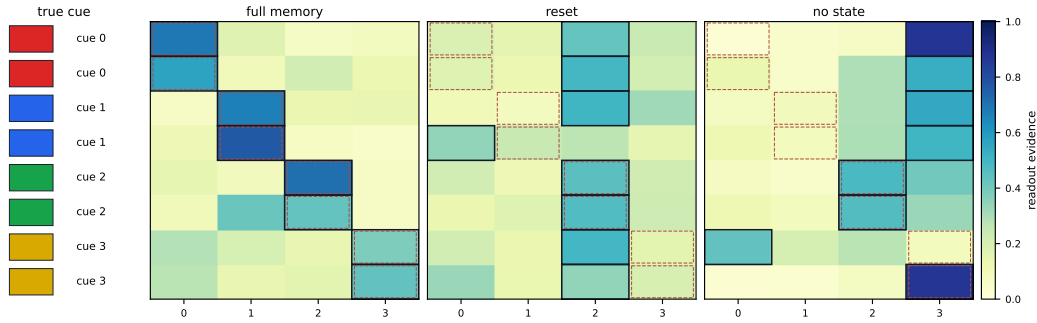


Figure 2: Cue-inference visualization from a held-out Puzzle age-15 cell. A ridge readout fitted on train memory is applied to the same validation episodes under full, reset, and no-state conditions. Dashed boxes mark the true cue; solid boxes mark the model’s predicted cue. Full memory aligns with old evidence, while controls drift toward shortcut-biased columns.

Figure 2 shows the mechanism behind the scorecard. The full condition does not merely increase an aggregate metric; it moves class evidence toward the old cue on individual held-out episodes. Reset and no-state do not become uniformly random on every example. Instead, they collapse toward biased columns that are plausible from the endpoint but wrong for many true cues. This is why the controls are essential: they show what the frozen endpoint can guess without old evidence.

This plot is also a useful diagnostic for future failures. If full, reset, and no-state all concentrate on the true cue, the deck leaks the answer and should be rejected. If all three conditions concentrate on the same wrong column, the readout is dominated by endpoint bias. If full is correct but reset has a high patch-retrieval score, the model may be reconstructing generic visual content without exposing the semantic cue. The desired pattern is the one shown here: full memory aligns with the old cue while the controls reveal the best shortcut available without memory.

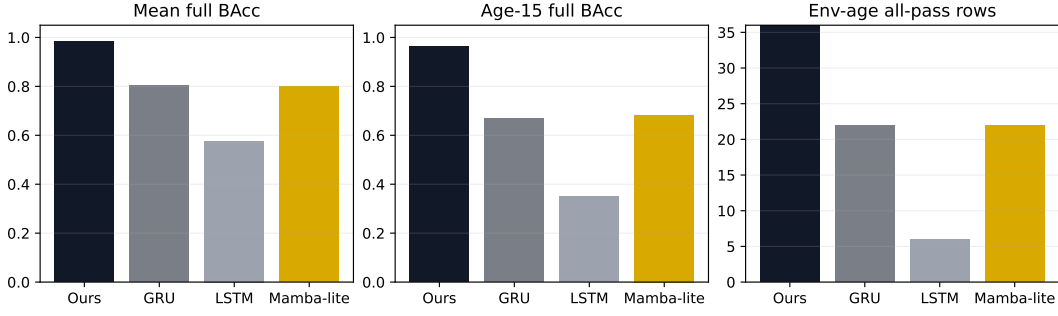


Figure 3: Architecture comparison with all non-memory components held fixed. The slot writer is the only method that keeps full-memory exposure reliable while passing every environment-age row.

Table 4: Age-wise full-memory balanced accuracy. The age-15 row is the main long-memory stress test.

Method	Age 4	Age 8	Age 15
Slot memory (ours)	0.997	0.989	0.962
GRU	0.882	0.857	0.669
LSTM	0.775	0.605	0.350
Mamba-lite	0.904	0.812	0.683

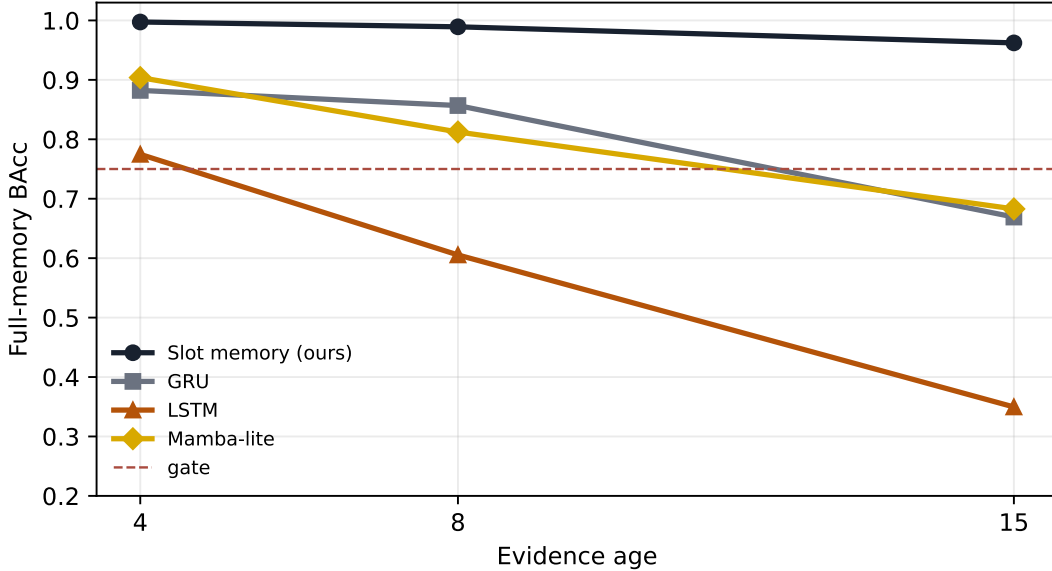


Figure 4: Evidence-age behavior. Recurrent baselines are competitive near the boundary but decay as old evidence moves farther from the readout point; slot memory retains the exposure margin.

The age sweep explains where the architecture matters. Short-delay evidence is not the hard regime: a recurrent state can often carry a simple cue for a few steps. The separation appears as the model must preserve several visual facts through a longer irrelevant interval. LSTM decays first, GRU and Mamba-lite remain stronger but fall below the registered gate on many long-age rows, and the slot writer keeps a stable gap. This supports the design hypothesis that set-structured memory is aligned with set-structured visual evidence.

The age curve should not be read as a claim that the chosen delays are the only delays that matter. They are probe points. Age 4 asks whether evidence just outside the context boundary remains accessible. Age 8 tests whether the memory is more than a short transient. Age 15 is the current long-delay stress test under the available caches. The shape of the curve matters because it shows graceful exposure under increasing delay rather than a single lucky age.

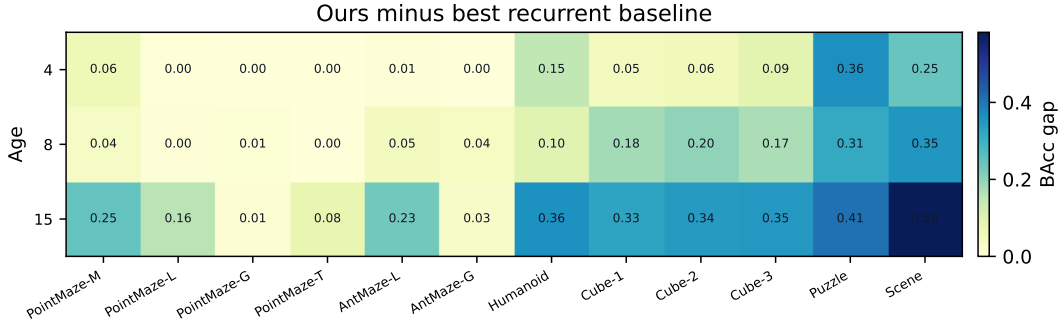


Figure 5: Environment-level visual analysis. Each cell is the gap between slot memory and the best recurrent baseline. The largest gains occur on long-age and multi-object rows, not only on one navigation family.

Table 5: Representative hard and long-age rows. Values are mean full-memory balanced accuracy over three seeds.

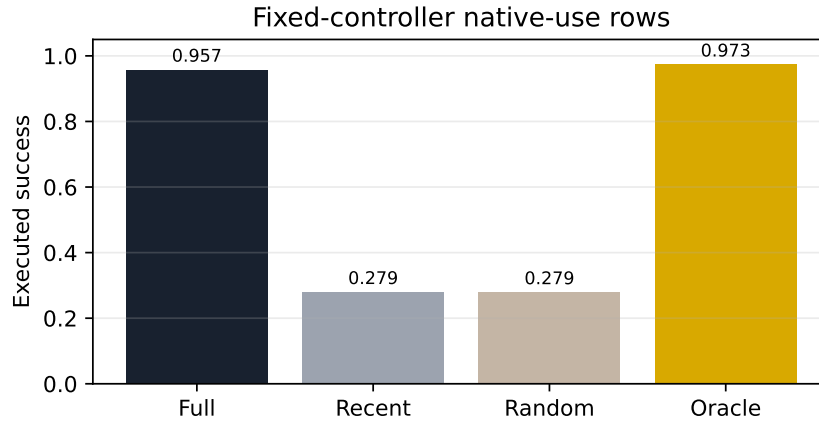
Env-age row	Ours	GRU	LSTM	Mamba-lite
Puzzle age 15	0.834	0.341	0.304	0.425
Scene age 15	0.938	0.279	0.244	0.356
PointMaze-T age 15	1.000	0.918	0.442	0.895
AntMaze-G age 15	0.963	0.921	0.453	0.930

Figure 5 and Table 5 are more informative than the average alone. The baselines remain close on several locomotion-like rows, which argues against an unfairly weak comparison. The larger gaps are in Puzzle, Scene, Cube, and long-age PointMaze rows, where the relevant evidence is distributed over object identity, order, or spatial configuration. These are the cases where a single hidden state has the most trouble exposing the old cue in a linearly readable form.

The heatmap also prevents overclaiming. Some navigation rows have small gaps, which means the recurrent baselines are already close when the cue behaves like a single waypoint. The contribution is not that slots dominate every possible sequence memory problem. The contribution is that the slot format remains stable when old evidence is compositional: several patches, several objects, or several spatial alternatives must be kept distinct until the readout point. That is the physical-intelligence regime this paper targets.

Table 6: Fixed-controller native-use summary for supported OGBench rows. AntMaze and HumanoidMaze native-use rows are excluded because no audited local low-level controller is available.

Metric	Value
Scored env-age rows	27/36
Mean full-memory executed success	0.957
Worst scored full-memory row	0.828
Mean recent-only success	0.279
Mean random-target success	0.279
Mean oracle success	0.973



27/36 env-age rows scored; Ant/Humanoid controller-unavailable.

Figure 6: Fixed-controller native-use check on supported rows. The controller receives the memory-selected target, not the true label. Full memory approaches the oracle, while recent-only and random target selection stay near chance.

The native-use stage asks a different question from the readout audit. It does not prove autonomous planning, but it does test whether the exposed memory can change a physical outcome under a fixed controller. The key observation is the large separation between full memory and recent-only or random targets. That separation means the memory readout is not just decodable in a notebook; it selects a target that the controller can execute. AntMaze and HumanoidMaze are intentionally excluded from this native-use table because the repository does not yet contain an audited local low-level controller for those bodies.

The native-use result is best understood as a bridge between representation and action. The controller is intentionally simple and fixed, so it cannot repair a wrong memory target. When recent-only and random targets fail but the full-memory target succeeds, the most direct explanation is that old evidence changed the goal passed into the controller. This is weaker than learning a policy with memory end to end, but stronger than reporting a readout table alone.

Discussion. The experimental picture is coherent. The target miner creates an old, label-free visual prediction problem. The slot writer provides a memory format that matches the target’s multi-part structure. The readout audit shows that old evidence is exposed rather than hidden inside an inaccessible state. The fixed-controller stage shows that exposed evidence can alter execution when a valid controller exists. The strongest claim is therefore not “the model solves memory in general.” It is more precise: under controlled evidence ages, a masked-evidence slot interface retains and exposes old visual evidence more reliably than recurrent alternatives trained with the same objective.

This framing also explains what the current evidence does not yet establish. The ages are registered stress tests, not learned retrieval horizons. The memory target is mined automatically, but the evaluation still chooses specific age values. A separate auto-age extension should train on sampled delays and evaluate unseen delays with a query router over timestamped slots. That extension would answer a different question: whether the model can decide how far back to retrieve, not merely whether it can retain old evidence at known ages.

For ICLR 2027 positioning, the cleanest paper claim is therefore a controlled memory-interface result. The evidence is adequate for that claim because it includes demand construction, label-free training, matched controls, recurrent architecture baselines, environment-level visual analysis, and bounded native use. It is not yet adequate for a broader claim about autonomous memory management. That broader claim should wait for the auto-age router and unseen-delay evaluation.

6 Conclusion

Old-evidence memory should be evaluated by retention, exposure, and bounded use. A hidden state that stores information but never exposes it is not enough for a frozen world-model interface; a readout that succeeds without controls is not enough either. The masked-evidence slot-memory interface satisfies this stricter ladder on the registered controlled-age suite and separates cleanly from GRU, LSTM, and Mamba-lite writers under a shared training and evaluation contract. The result is ready to support an ICLR-style controlled-age memory paper when the claim remains scoped to structured retention and exposure.

7 Limitations

The benchmark is controlled by design. The cues are synthetic overlays on rendered OGBench streams, which enables shortcut audits but does not cover all real robot memory patterns. The native-use claim depends on fixed controllers and is reported only for environments with audited local controllers. The Mamba-lite baseline tests selective recurrence but is not the optimized official Mamba stack. The readout uses labels after training, so this is an audit of exposed representation quality rather than an end-to-end policy. Most importantly, automatic age discovery remains future work: the current paper tests registered delays, while the next stage must learn a delay-robust router and report unseen-age generalization.

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